Using Technology Safely

Importance of physical security

- Hundreds of thousands of phones are lost or stolen every year - it’s easily done!
- Your phone gives you access to almost everything online (including email!) and many of our accounts are left logged-in on our phones
- Your phone is likely to be your second factor! Someone with your phone could use it to get your 2-factor-authentication (2FA) codes.

Lock types: PINs

- PIN is the most common/simple method
- PINs don’t have to be as strong as website passwords as they are kept on the device
- Most smartphones will add a ‘cooldown’ period to stop repeated attempts
- BUT they can be **seen** or **guessed** if you aren’t careful

Attempts	3	4	5	6	7	8	9	10 or more
iOS and iPadOS Lock Screen	None	1 minute	5 minutes	15 minutes	1 hour	3 hours	8 hours	Device is disabled and must connect to a Mac or PC

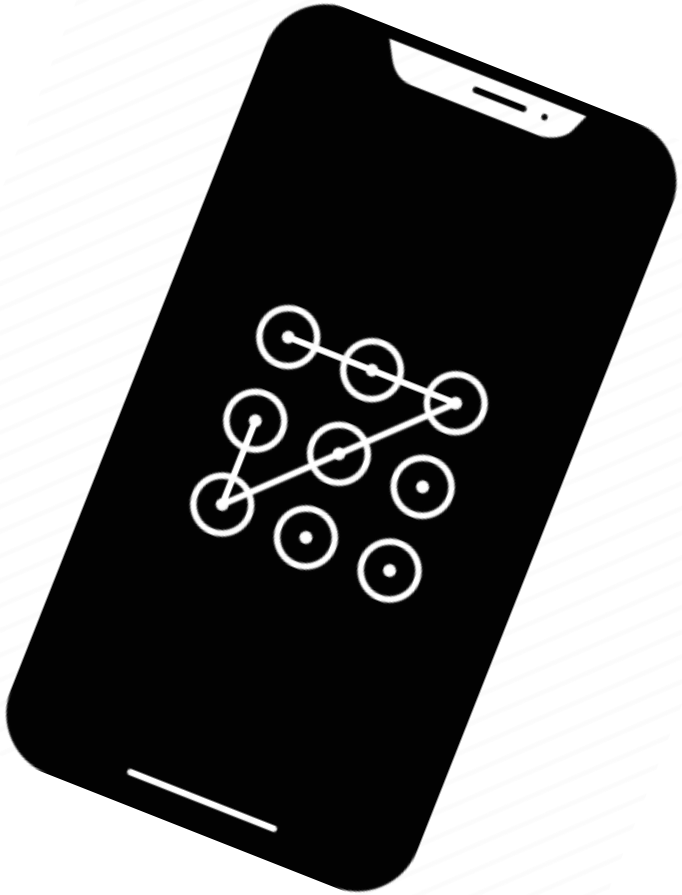
Good habits for PINs

- **Avoid** simple or obvious patterns
  - Repeated digits
  - Sequential numbers
- Don’t use personal information
- Use 6 digits instead of 4
  - Far stronger (assuming it’s not predictable)
  - 100x as many possibilities
  - Harder to spot by **shoulder surfing**

Pattern locks

- A 3x3 grid of dots where you draw a line between the dots, forming a pattern
- You might find this easier to remember
- Research tells us 2 ways to make it harder for shoulder-surfers to memorise our pattern:

1. A longer, more complex pattern
2. Disable the ‘guide lines’ if your phone settings allow



...and clean your screen!

PINs and (especially) patterns can be guessed (or at least narrowed down) based on smudges on the screen.

Biometrics

- Finger scanner, face scanner… voice recognition?
- Usually more convenient
- No worries about ‘shoulder surfing’
- Template stored on mobile device – not sent to ‘the cloud’
  - You might need to rescan your fingerprints occasionally as they change over time
- Can sometimes be inconsistent – you still need another unlock method, such as a PIN or pattern

Reality check – are they secure?

- Good news – they're all quite strong!
- It's very unlikely someone can get into a locked phone unless they already know the password – so be aware of your surroundings for shoulder-surfers!
- The software and tools that could help someone 'crack' a phone are very exclusive – they are expensive and often only sold to law enforcement

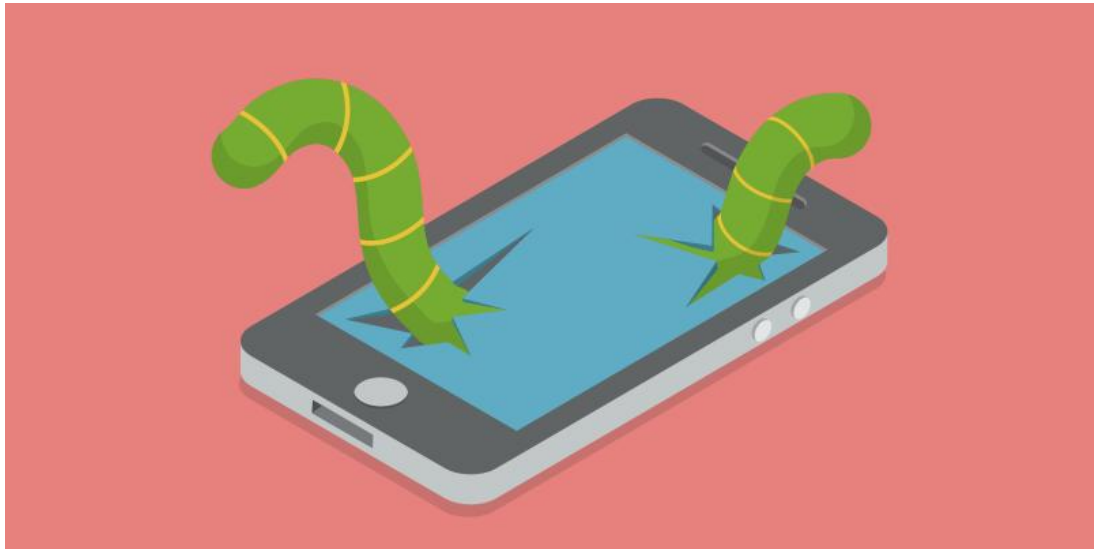
Viruses and Malware

- **Malware** is the overarching term for **malicious software**
- A virus is a type of malware, but there are others
- You can get them from
  - Downloaded software
  - Emails
  - Websites
  - USB drives
- Be very careful installing software:
  - Especially if someone else has asked you to (scammers often will)
- On mobile, always install from the app store for your device
  - Apple: App Store
  - Google: Google Play

Malware: Virus/Worms

- What do they do?
  - Just like a real virus:
  - Multiply and spread across files! Worms do this all on their own
  - Delete files
  - Can install ‘backdoors’ (send information to attacker)
  - Can just slow down your computer

- Alarm bells:
  - Computer running slowly
  - Files are missing
  - Your hard drive space is disappearing



Malware: Trojan (horse)

- What do they do?
  - Does everything a virus/worm does
    - Delete files
    - Download other malware
    - Remote access
    - Steal data (spyware)
  - Needs to be ‘activated’ by the user
  - Usually comes bundled with another (real) program
  - Remember your vigilance tips!

- Alarm bells:
  - Computer running slowly
  - Files are missing
  - Your hard drive space is disappearing

You can buy trojan ‘kits’ on the internet which allow you to make simple variations of existing trojans.

Usually spread through email or messaging applications (users will download them)

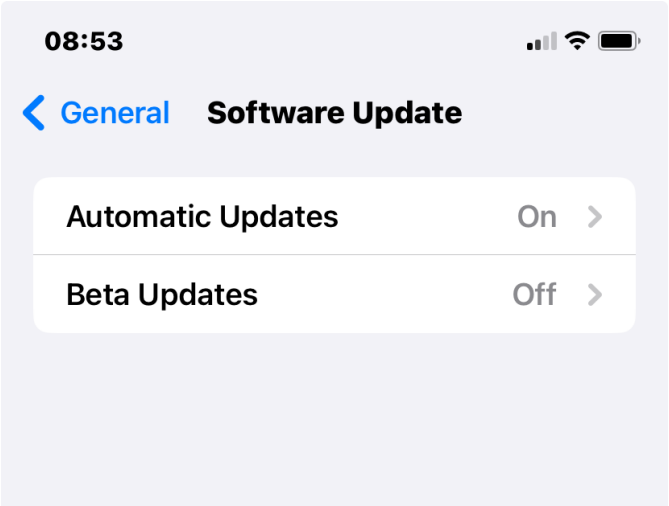
This means that it is very difficult for antivirus software to keep up to date!

Malware: Ransomware

- What do they do?
  - It is a form of trojan
  - It will encrypt your files so you can no longer use them!
- Alarm bells:
  - You get a message saying your files have been encrypted
- Do not:
  - Pay the ransom. Do you trust them to decrypt your files?
  - Delete the ransom note! You might be able to decrypt your files if you know how they were encrypted
    - https://id-ransomware.malwarehunterteam.com/
- Do:
  - Disconnect the infected computer from the internet – ransomware can spread!

Software updates

- Your devices are run by **software**
  - Made by people
  - People make mistakes
- When software vulnerabilities are discovered, updates will be released to “patch” those issues
- **Always keep your devices up to date**
  - Consider enabling automatic updates in your settings
- Be wary of using mobile devices that can no longer receive security updates



Antivirus/antimalware software on computers

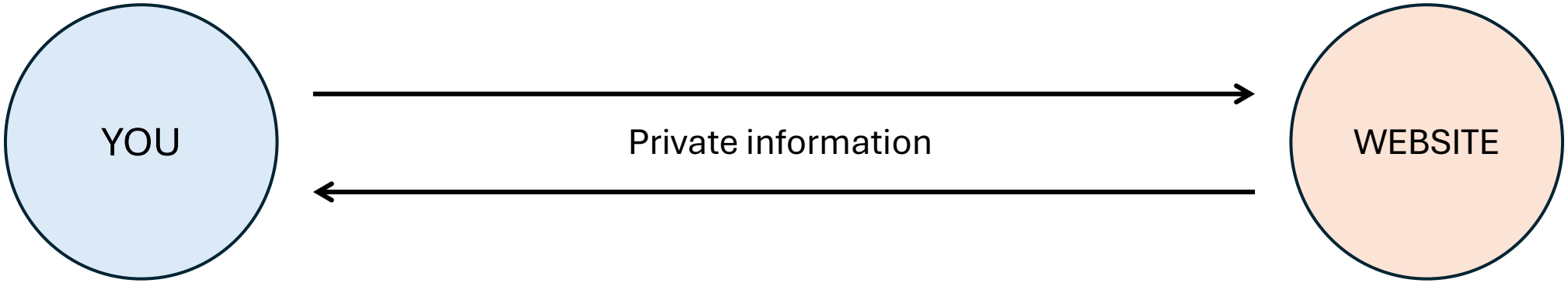
- Apple and Windows computers have adequate antivirus software included for free.
- Additional antivirus software exists, and you might get a free trial with your computer
  - You might not need anything additional to the built in software
  - Beware being upsold additional things you might not use
  - 3rd party antivirus software might interfere with your built-in protection
  - Research these products before installing them
- Make sure your antivirus software is kept updated automatically
- Set your software to scan new files
  - Run a full system scan if you're worried about your computer

Antivirus/antimalware software on mobile devices

- No additional software should be needed
- Remember, only install from the official app stores
- Keep your device and apps updated automatically

Browsing safely

Man in the middle attacks



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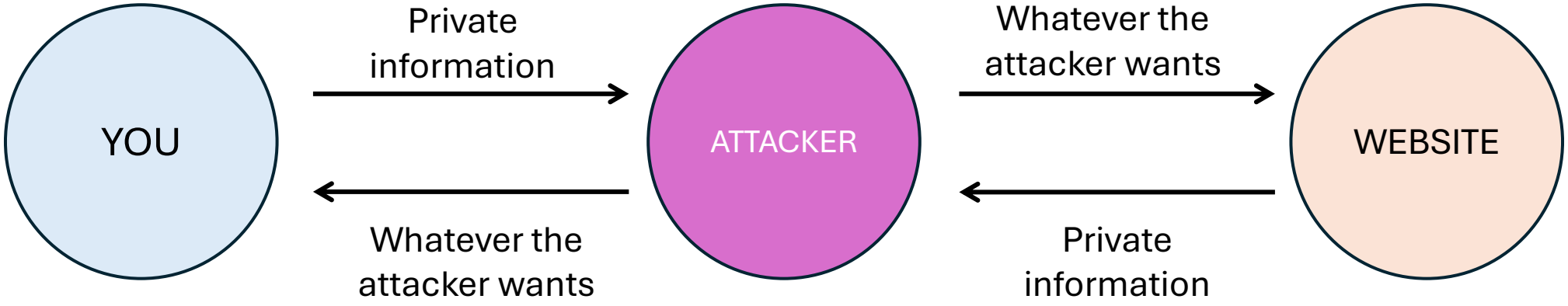

Man in the middle attack types

- On the same WiFi network as you and pretends to be your router
- Setting up a fake WiFi network
  - e.g. Linskill_Free_FAST
- Compromised WiFi router
  - e.g. default weak password
- Unsafe apps or Malware on your device

Avoiding Man In The Middle: HTTPS

- Websites that start with https:// not http://
  - This is most websites, now
- Connections to HTTPS websites are encrypted
  - It is gibberish to anyone who intercepts
- A website having HTTPS doesn't mean it's safe, only that other people can't see what is sent.
  - Scam websites will probably have https too
- HTTPS connections are backed up by certificates
  - You can check that they are who they say they are on your web browser

Avoiding Man In The Middle: Public WiFi

- Public WiFi network introduce risk
- If possible, use mobile data instead
  - Also known as “hotspot” on your device
- If you need to use public WiFi, limit activities
  - You don’t know who is controlling the WiFi
- If you use public WiFi, double check that the network is correct
  - E.g. check with reception what their WiFi is called

Avoiding Man In The Middle: VPNs

- Virtual Private Network (VPN)
- Creates a secure connection with another computer somewhere else (usually another country)
- Everything you do goes through that computer and can’t be intercepted
- However
  - The VPN service will see everything you do
  - VPNs usually cost money
  - Free VPNs could be malicious
  - VPNs can slow down your computer
  - Probably overkill in many cases

URLs

- URLs are **web addresses**
  - For example, https://www.bbc.co.uk/news
- They are **unique**
- They can be confusing to look at
- Scammers can create URLs aiming to trick users
  - Websites that are very close to the real thing but one letter off
  - Websites that switch parts of the URL around
  - Email addresses that look like the real deal

https://www.bbc.co.uk/news

- https tells us the connection is encrypted
- www = world wide web
  - This part doesn't really matter any more
- bbc.co.uk is the **domain**, the actual website you're visiting
- /news is a specific page on that site
- Websites always have a top-level domain
  - .com, .co.uk, .org, .ac.uk etc
- The part just before the top-level domain is the name of the website.
  - **Read from the right**

Always look carefully!

bbc.co.uk.breaking-news.com

- Goes to breaking-news.com (read from the right)

bbc-news.co.uk

- Does go to bbc-news, but this is not the domain used by the BBC
- Search engines can (usually) help you find the right domain
- Password managers can be useful here, as they remember the URL for your accounts

bbc.co.uk.info

- Goes to uk.info (.info is one of the rarer top-level domains)

bbc.co.uk/news

- Goes to the news page at the BBC website

The same applies to emails

Sender:

- northumbria@james.com
  - This is from account “Northumbria” at james.com
- james@northumbria-direct.com
  - Northumbria-direct isn’t the right domain
- jamesnorthumbria@gmail.com
  - Anyone can make a gmail account (or any other widely available email service)
- More on phishing scams tomorrow …

QR codes and shortened links

- QR codes are two dimensional barcodes
- They express data visually, usually URLs
- They can be genuinely very useful
- BUT, use them with caution
  - Check the link
  - Watch out for fake QR codes stuck over genuine ones (e.g. parking meters)
- Similarly, avoid shortened links where possible (e.g. tinyurl.com/jf25cac2)
  - Could go anywhere

Safe shopping

- Check the URL
- Look for HTTPS
- Stick to retailers you know (where possible)
  - Check reviews for unfamiliar sites (e.g. trustpilot)
  - Avoid following ads on social media
- Be wary of amazing deals
- Credit card offers good fraud protection
- PayPal / Apple Pay / Google Wallet can prevent you needing to enter card details

Cookies

- Cookes are small pieces of text that get saved to your device
- Websites use them to remember things about you
  - Logging in; remembering shopping cart etc
  - These things are usually classes as “essential”
- They can also be used to record other things
  - Websites you’ve visited
  - What you’ve looked at
- Third party cookies let advertisers track you across the internet
- Not a security risk (usually), but damages privacy
- In the EU (and UK), websites have to tell you what cookies they use and let you object to non-essential cookies
  - Look for “reject all”, “manage preferences” or “essential only”
- Some browsers (or browser extensions) let you control what cookies you allow

Key Takeaways

- Keep your devices safe and secure with a strong PIN
- Keep your devices up to date
- Make sure your device is protected
  - Mobile devices – always use official app stores
- Avoid public WiFi where you can
- Know how to read URLs and email addresses
  - Be cautious when scanning QR codes
- Shop sensibly and with known retailers
  - Otherwise research carefully and take precautions
- Don’t worry about cookies, but stick to essential where you can